

Recitation 4: Simplification and Karnaugh maps

1 Theorems

Axioms

1a $0 \cdot 0 = 0$

1b $1 + 1 = 1$

2a $1 \cdot 1 = 1$

2b $0 + 0 = 0$

3a $0 \cdot 1 = 1 \cdot 0 = 0$

3b $1 + 0 = 0 + 1 = 1$

4a if $x = 0$, then $\bar{x} = 1$

4b if $x = 1$, then $\bar{x} = 0$

Single-variable theorems

5a $x \cdot 0 = 0$

5b $x + 1 = 1$

6a $x \cdot 1 = x$

6b $x + 0 = x$

7a $x \cdot x = x$

7b $x + x = x$

8a $x \cdot \bar{x} = 0$

8b $x + \bar{x} = 1$

9 $\bar{\bar{x}} = x$

Two- and three-variable theorems

10a $x \cdot y = y \cdot x$

Commutative

14a $xy + x\bar{y} = x$

Combining

10b $x + y = y + x$

14b $(x + y)(x + \bar{y}) = x$

11a $x(yz) = (xy)z$

Associative

15a $\overline{xy} = \bar{x} + \bar{y}$

DeMorgan

11b $x + (y + z) = (x + y) + z$

15b $\overline{x + y} = \bar{x} \bar{y}$

12a $x(y + z) = xy + xz$

Distributive

16a $x + \bar{x}y = x + y$

Redundant literal

12b $x + yz = (x + y)(x + z)$

16b $x(\bar{x} + y) = xy$

13a $x + xy = x$

Absorption

17a $xy + \bar{x}z + yz = xy + \bar{x}z$

Consensus

13b $x(x + y) = x$

17b $(x + y)(y + z)(\bar{x} + z) = (x + y)(\bar{x} + z)$

Duality. Swap $\cdot$ with $+$, and 0 with 1, and a true identity becomes another true identity. x , y and z can each stand for a whole expression: in $AB + \bar{A}\bar{B}C$, x is AB .

2 Combining

$$xy + x\bar{y} = x \quad (x + y)(x + \bar{y}) = x$$

Problem 1

Simplify. Name the theorem at each step.

a. $F = \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}C + ABC$

b. $F = (A + B + C)(A + B + \bar{C})$

c. $F = \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}CD + \bar{A}B\bar{C}D + \bar{A}BCD$

3 Absorption

$$x + xy = x \qquad x(x + y) = x$$

Problem 2

Simplify. Name the theorem at each step.

- a. $F = \overline{A}B + \overline{A}B\overline{C}D + \overline{A}BC\overline{D}$
- b. $F = (A + B)(A + B + C)$
- c. $F = B + AB + BC + ABC$

4 Redundant literal

$$x + \overline{x}y = x + y \qquad x(\overline{x} + y) = xy$$

Problem 3

Simplify. Name the theorem at each step.

- a. $F = C + \overline{A}B\overline{C}$
- b. $F = \overline{A} + AB\overline{C}$
- c. $F = B(A + \overline{B})$
- d. $F = A + \overline{A}B + \overline{A}\overline{B}C$

5 Consensus

$$xy + \overline{x}z + yz = xy + \overline{x}z \qquad (x + y)(y + z)(\overline{x} + z) = (x + y)(\overline{x} + z)$$

Problem 4

Find the redundant term and delete it.

- a. $F = \overline{A}\overline{B} + AC + \overline{B}C$
- b. $F = A\overline{B}C + \overline{A}D + \overline{B}CD$
- c. $F = AB + B\overline{C} + AC$
- d. $F = (A + B)(\overline{A} + C)(B + C)$

6 DeMorgan

$$\overline{xy} = \overline{x} + \overline{y} \quad \overline{x + y} = \overline{x} \overline{y}$$

Problem 5

Remove every bar that covers more than one letter, then simplify.

- a. $F = \overline{\overline{AB} + \overline{C}} + AC$
- b. $F = \overline{\overline{A} + \overline{BC}}$
- c. $F = \overline{A + B} + \overline{A + B}$

7 Brackets: the distributive laws

$$x(y + z) = xy + xz \quad x + yz = (x + y)(x + z)$$

Problem 6

Simplify. Name the theorem at each step.

- a. $F = (A + BC)(A + \overline{BC})$
- b. $F = (A + \overline{B})(\overline{A} + C)$
- c. $F = (A + B + C)(A + B + D)$

8 Using a term twice

$$x = x + x$$

Problem 7

Simplify. Say which term you copy.

- a. $F = \overline{A} \overline{B} \overline{C} + \overline{A} \overline{B} C + \overline{A} B C$
- b. $F = A \overline{B} \overline{C} + A \overline{B} C + A B C$
- c. $F = \overline{A} B \overline{C} D + \overline{A} B C D + A B C D$

9 Karnaugh maps

Each cell is one row of the truth table; m_5 is the row $ABC = 101$. Rows and columns go 00, 01, 11, 10, so neighbouring cells differ in one variable, and the map wraps around its edges. Circle the 1s in groups of 1, 2, 4 or 8, as large as possible, and keep only the variables that do not change inside a group.

		BC			
		00	01	11	10
A	0	m_0	m_1	m_3	m_2
	1	m_4	m_5	m_7	m_6

		CD			
		00	01	11	10
AB	00	m_0	m_1	m_3	m_2
	01	m_4	m_5	m_7	m_6
	11	m_{12}	m_{13}	m_{15}	m_{14}
	10	m_8	m_9	m_{11}	m_{10}

Problem 8

Fill in the map, circle the groups, and write the simplest SOP.

a. $F = \Sigma m(3, 5, 6, 7)$

		BC			
		00	01	11	10
A	0	0	1	3	2
	1	4	5	7	6

b. $F = \Sigma m(0, 1, 3)$

		BC			
		00	01	11	10
A	0	0	1	3	2
	1	4	5	7	6

c. $F = \Sigma m(0, 2, 4, 5, 6)$

		BC			
		00	01	11	10
A	0	0	1	3	2
	1	4	5	7	6

Problem 9

Fill in the map, circle the groups, and write the simplest SOP.

a. $F = \Sigma m(0, 2, 5, 7, 8, 10, 13, 15)$

		CD			
		00	01	11	10
AB	00	0	1	3	2
	01	4	5	7	6
	11	12	13	15	14
	10	8	9	11	10

b. $F = \Sigma m(1, 3, 4, 5, 6, 7, 9, 11, 13, 15)$

		CD			
		00	01	11	10
AB	00	0	1	3	2
	01	4	5	7	6
	11	12	13	15	14
	10	8	9	11	10

c. $F = \Sigma m(0, 1, 2, 3, 4, 6, 8, 9, 10, 11)$

		CD			
		00	01	11	10
AB	00	0	1	3	2
	01	4	5	7	6
	11	12	13	15	14
	10	8	9	11	10

Solutions

Problem 1

a. $F = \overline{A}\overline{B}C + \overline{A}BC + A\overline{B}C + ABC$

$$= \overline{A}C + AC$$

$xy + x\overline{y} = x$ on each pair (they differ only in B)

$$= C$$

$xy + x\overline{y} = x$ (they differ only in A)

b. $F = (A + B + C)(A + B + \overline{C})$

$$= A + B$$

$(x + y)(x + \overline{y}) = x$ with $x = A + B$, $y = C$

c. $F = \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}CD + \overline{A}B\overline{C}D + \overline{A}BCD$

$$= \overline{A}\overline{B}D + \overline{A}BD$$

$xy + x\overline{y} = x$ on each pair (they differ only in C)

$$= \overline{A}D$$

$xy + x\overline{y} = x$ (they differ only in B)

Problem 2

a. $F = \overline{A}B + \overline{A}B\overline{C}D + \overline{A}BC\overline{D}$

$$= \overline{A}B + \overline{A}BC\overline{D}$$

$x + xy = x$: $\overline{A}B\overline{C}D$ contains $\overline{A}B$

$$= \overline{A}B$$

$x + xy = x$: $\overline{A}BC\overline{D}$ contains $\overline{A}B$

b. $F = (A + B)(A + B + C)$

$$= A + B$$

$x(x + y) = x$ with $x = A + B$, $y = C$

c. $F = B + AB + BC + ABC$

$$= B + BC + ABC$$

$x + xy = x$: AB contains B

$$= B + ABC$$

$x + xy = x$: BC contains B

$$= B$$

$x + xy = x$: ABC contains B

Problem 3

a. $F = C + \overline{A}B\overline{C}$

$$= C + \overline{A}B$$

$x + \overline{x}y = x + y$ with $x = C$

b. $F = \overline{A} + AB\overline{C}$

$$= \overline{A} + B\overline{C}$$

$x + \overline{x}y = x + y$ with $x = \overline{A}$, so $\overline{x} = A$

c. $F = B(A + \overline{B})$

$$= AB$$

$x(\overline{x} + y) = xy$ with $x = B$

$$\begin{aligned}
\text{d. } F &= A + \overline{A}B + \overline{A}\overline{B}C \\
&= A + B + \overline{A}\overline{B}C & x + \overline{x}y &= x + y \text{ with } x = A \\
&= A + B + \overline{A}C & x + \overline{x}y &= x + y \text{ with } x = B \\
&= A + B + C & x + \overline{x}y &= x + y \text{ with } x = A
\end{aligned}$$

Problem 4

$$\begin{aligned}
\text{a. } F &= \overline{A}\overline{B} + AC + \overline{B}C & \overline{A}\overline{B} \text{ and } AC \text{ (} x = A \text{) give the consensus term } \overline{B} \cdot C &= \overline{B}C \\
&= \overline{A}\overline{B} + AC & xy + \overline{x}z + yz &= xy + \overline{x}z: \text{ delete } \overline{B}C \\
\\
\text{b. } F &= A\overline{B}C + \overline{A}D + \overline{B}CD & A\overline{B}C \text{ and } \overline{A}D \text{ (} x = A \text{) give the consensus term } \overline{B}C \cdot D &= \overline{B}CD \\
&= A\overline{B}C + \overline{A}D & xy + \overline{x}z + yz &= xy + \overline{x}z: \text{ delete } \overline{B}CD \\
\\
\text{c. } F &= AB + B\overline{C} + AC & B\overline{C} \text{ and } AC \text{ (} x = C \text{) give the consensus term } B \cdot A &= AB \\
&= B\overline{C} + AC & xy + \overline{x}z + yz &= xy + \overline{x}z: \text{ delete } AB \\
\\
\text{d. } F &= (A + B)(\overline{A} + C)(B + C) & A + B \text{ and } \overline{A} + C \text{ (} x = A \text{) give the consensus term } B + C \\
&= (A + B)(\overline{A} + C) & (x + y)(y + z)(\overline{x} + z) &= (x + y)(\overline{x} + z)
\end{aligned}$$

Problem 5

$$\begin{aligned}
\text{a. } F &= \overline{\overline{A}\overline{B} + \overline{C} + AC} \\
&= \overline{\overline{A}\overline{B}} \cdot \overline{\overline{C}} + AC & \overline{x + y} &= \overline{x} \cdot \overline{y} \\
&= \overline{ABC} + AC & \overline{\overline{x}} &= x \\
&= AC & x + xy &= x: ABC \text{ contains } AC \\
\\
\text{b. } F &= \overline{\overline{A} + \overline{B}\overline{C}} \\
&= \overline{\overline{A}} \cdot \overline{\overline{B}\overline{C}} & \overline{x + y} &= \overline{x} \cdot \overline{y} \\
&= \overline{A}\overline{\overline{B}\overline{C}} & \overline{\overline{x}} &= x \\
&= A(\overline{B} + \overline{\overline{C}}) & \overline{xy} &= \overline{x} + \overline{y} \\
&= A(\overline{B} + C) & \overline{\overline{x}} &= x \\
&= A\overline{B} + AC & x(y + z) &= xy + xz \\
\\
\text{c. } F &= \overline{\overline{A} + \overline{B}} + \overline{\overline{A} + \overline{B}} \\
&= \overline{A}\overline{B} + \overline{\overline{A}\overline{B}} & \overline{x + y} &= \overline{x} \cdot \overline{y} \text{ on each term} \\
&= \overline{A}\overline{B} + \overline{A}B & \overline{\overline{x}} &= x \\
&= \overline{A} & xy + x\overline{y} &= x \text{ (they differ only in } B \text{)}
\end{aligned}$$

Problem 6

a. $F = (A + BC)(A + \overline{BC})$

$$= A + (BC)(\overline{BC}) \quad x + yz = (x + y)(x + z) \text{ with } x = A$$

$$= A + B\overline{BC} \quad xy = yx \text{ and } x \cdot x = x \text{ } (C \cdot C = C)$$

$$= A + 0 \cdot C \quad x \cdot \overline{x} = 0$$

$$= A + 0 \quad x \cdot 0 = 0$$

$$= A \quad x + 0 = x$$

b. $F = (A + \overline{B})(\overline{A} + C)$

$$= A\overline{A} + AC + \overline{A}\overline{B} + \overline{BC} \quad x(y + z) = xy + xz$$

$$= 0 + AC + \overline{A}\overline{B} + \overline{BC} \quad x \cdot \overline{x} = 0$$

$$= AC + \overline{A}\overline{B} + \overline{BC} \quad x + 0 = x$$

$$= AC + \overline{A}\overline{B} \quad xy + \overline{x}z + yz = xy + \overline{x}z: \overline{BC} \text{ is the consensus of } AC \text{ and } \overline{A}\overline{B}$$

c. $F = (A + B + C)(A + B + D)$

$$= A + B + CD \quad x + yz = (x + y)(x + z) \text{ with } x = A + B$$

Problem 7

a. $F = \overline{A}\overline{B}\overline{C} + \overline{A}\overline{BC} + \overline{ABC}$

$$= \overline{A}\overline{B}\overline{C} + \overline{A}\overline{BC} + \overline{A}\overline{BC} + \overline{ABC} \quad x = x + x: \text{ write } \overline{A}\overline{BC} \text{ twice}$$

$$= (\overline{A}\overline{B}\overline{C} + \overline{A}\overline{BC}) + (\overline{A}\overline{BC} + \overline{ABC}) \quad \text{pair each copy with one neighbour}$$

$$= \overline{A}\overline{B} + \overline{AC} \quad xy + x\overline{y} = x \text{ on each pair}$$

b. $F = \overline{A}\overline{B}\overline{C} + \overline{A}\overline{BC} + \overline{ABC}$

$$= \overline{A}\overline{B}\overline{C} + \overline{A}\overline{BC} + \overline{A}\overline{BC} + \overline{ABC} \quad x = x + x: \text{ write } \overline{A}\overline{BC} \text{ twice}$$

$$= (\overline{A}\overline{B}\overline{C} + \overline{A}\overline{BC}) + (\overline{A}\overline{BC} + \overline{ABC}) \quad \text{pair each copy with one neighbour}$$

$$= \overline{A}\overline{B} + \overline{AC} \quad xy + x\overline{y} = x \text{ on each pair}$$

c. $F = \overline{A}\overline{B}\overline{C}D + \overline{A}BCD + ABCD$

$$= \overline{A}\overline{B}\overline{C}D + \overline{A}BCD + \overline{A}BCD + ABCD \quad x = x + x: \text{ write } \overline{A}BCD \text{ twice}$$

$$= (\overline{A}\overline{B}\overline{C}D + \overline{A}BCD) + (\overline{A}BCD + ABCD) \quad \text{pair each copy with one neighbour}$$

$$= \overline{A}BD + BCD \quad xy + x\overline{y} = x \text{ on each pair}$$

Problem 8

a. $F = \Sigma m(3, 5, 6, 7)$

		BC			
		00	01	11	10
A	0	0	0	1	0
	1	0	1	1	1

BC : m_3, m_7 ; A changes

AC : m_5, m_7 ; B changes

AB : m_6, m_7 ; C changes

$$F = AB + BC + AC$$

b. $F = \Sigma m(0, 1, 3)$

		BC			
		00	01	11	10
A	0	1	1	1	0
	1	0	0	0	0

$\overline{A}\overline{B}$: m_0, m_1 ; C changes

$\overline{A}C$: m_1, m_3 ; B changes

$$F = \overline{A}\overline{B} + \overline{A}C$$

c. $F = \Sigma m(0, 2, 4, 5, 6)$

		BC			
		00	01	11	10
A	0	1	0	0	1
	1	1	1	0	1

$\overline{C}$: m_0, m_2, m_4, m_6 (the edges touch); A, B change

$A\overline{B}$: m_4, m_5 ; C changes

$$F = \overline{C} + A\overline{B}$$

Problem 9

a. $F = \Sigma m(0, 2, 5, 7, 8, 10, 13, 15)$

		CD			
		00	01	11	10
AB	00	1	0	0	1
	01	0	1	1	0
	11	0	1	1	0
	10	1	0	0	1

$\overline{B}\overline{D}$: the four corners; A, C change

BD : m_5, m_7, m_{13}, m_{15} ; A, C change

$$F = \overline{B}\overline{D} + BD$$

b. $F = \Sigma m(1, 3, 4, 5, 6, 7, 9, 11, 13, 15)$

		CD			
		00	01	11	10
AB	00	0	1	1	0
	01	1	1	1	1
	11	0	1	1	0
	10	0	1	1	0

D : columns $CD = 01, 11$; A, B, C change

$\overline{A}B$: row $AB = 01$; C, D change

$$F = D + \overline{A}B$$

c. $F = \Sigma m(0, 1, 2, 3, 4, 6, 8, 9, 10, 11)$

		CD			
		00	01	11	10
AB	00	1	1	1	1
	01	1	0	0	1
	11	0	0	0	0
	10	1	1	1	1

$\overline{B}$: rows $AB = 00, 10$ (top and bottom touch); A, C, D change

$\overline{A}\overline{D}$: m_0, m_2, m_4, m_6 (the edges touch); B, C change

$$F = \overline{B} + \overline{A}\overline{D}$$